

FINITE-DIMENSIONAL BANACH TRANSFER FOR THE ALBERTI–MARCHESE FLAT-CHAIN THEOREM

RUN FA_BANACH_001

Status. `partial_result_likely_valid`. This packet records a finite-dimensional Banach-space subcase adjacent to the higher-dimensional Banach extension question noted in Bate–Caputo–Takac–Valentine–Wald, arXiv:2508.08017.

SOURCE TARGET

The source paper proves the 1-dimensional flat-chain conjecture and a Banach-space version of the Alberti–Marchese structure theorem for 1-currents. In its discussion of higher-dimensional currents, the authors state that for $k \geq 2$ the strict polyhedral approximation result fails in every infinite-dimensional Hilbert space and that it is therefore unclear whether Alberti–Marchese’s Theorem 1.1 can be extended beyond finite-dimensional Euclidean space.

The supporting Euclidean theorem is Alberti–Marchese, *On the structure of flat chains with finite mass*, arXiv:2311.06099, Theorem 1.1: every real k -dimensional flat chain with finite mass in $\mathbb{R}^d$, $1 \leq k < d$, is the restriction of a boundaryless normal k -current to a Borel set, with mass at most $(2 + \varepsilon)$ times the mass of the original chain.

PROOF INTUITION

Finite-dimensional Banach spaces are linearly bi-Lipschitz to Euclidean space. The structure theorem can therefore be transported across such an isomorphism: push the flat chain to $\mathbb{R}^d$, use Alberti–Marchese, and pull the resulting normal current back. The price is exactly the usual Lipschitz mass distortion: a k -current mass changes by at most the k -th power of the Lipschitz constant.

Partial theorem. *Let E be a d -dimensional Banach space, let $1 \leq k < d$, and let T be a real k -dimensional flat chain with finite mass in E . Let $A : E \rightarrow \mathbb{R}^d$ be a linear isomorphism, where $\mathbb{R}^d$ carries its Euclidean norm. Then for every $\varepsilon > 0$ there exist a boundaryless normal k -current N in E and a Borel set $B \subset E$ such that*

$$T = N \llcorner B$$

and

$$\mathbb{M}_E(N) \leq (2 + \varepsilon) \|A\|^k \|A^{-1}\|^k \mathbb{M}_E(T).$$

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Consequently the constant may be written with the k -th power of the Banach–Mazur distance from E to ℓ_2^d , up to arbitrarily small slack in the choice of A .

Proof. Because A is bi-Lipschitz, pushforward by A sends flat k -chains in E to flat k -chains in $\mathbb{R}^d$. Indeed, if P_i is a polyhedral flat-norm approximating sequence for T , then $A_{\#}P_i$ is a polyhedral approximating sequence for $A_{\#}T$, with flat distances controlled by the Lipschitz constants of A . Also

$$\mathbb{M}_{\mathbb{R}^d}(A_{\#}T) \leq \|A\|^k \mathbb{M}_E(T).$$

Apply Alberti–Marchese’s Euclidean theorem to $A_{\#}T$. For any $\delta > 0$ there are a boundaryless normal k -current N' in $\mathbb{R}^d$ and a Borel set $B' \subset \mathbb{R}^d$ such that

$$A_{\#}T = N' \llcorner B'$$

and

$$\mathbb{M}_{\mathbb{R}^d}(N') \leq (2 + \delta) \mathbb{M}_{\mathbb{R}^d}(A_{\#}T).$$

Set $N := (A^{-1})_{\#}N'$ and $B := A^{-1}(B')$. Pushforward commutes with boundary, so $\partial N = (A^{-1})_{\#}\partial N' = 0$, and N is normal because normality is preserved under Lipschitz pushforward. Restriction to a Borel set commutes with pushforward by the bijection A^{-1} , hence

$$N \llcorner B = (A^{-1})_{\#}(N' \llcorner B') = (A^{-1})_{\#}A_{\#}T = T.$$

Finally,

$$\mathbb{M}_E(N) \leq \|A^{-1}\|^k \mathbb{M}_{\mathbb{R}^d}(N') \leq (2 + \delta) \|A^{-1}\|^k \|A\|^k \mathbb{M}_E(T).$$

Taking $\delta = \varepsilon$ gives the displayed estimate. Taking the infimum over linear isomorphisms $A : E \rightarrow \ell_2^d$, with the usual small slack because the infimum need not be attained, gives the Banach–Mazur formulation. $\square$

VERIFICATION AND LIMITATIONS

The argument uses only standard functorial properties of metric currents and flat chains under Lipschitz maps, plus Alberti–Marchese’s Euclidean theorem. No computation is involved.

This is not a solution of the infinite-dimensional Banach problem mentioned in the source paper. It also does not prove the sharp $(2 + \varepsilon)$ mass constant for non-Euclidean finite-dimensional norms, and it does not identify arbitrary metric k -currents in Banach spaces with flat chains.

SEARCH NOTES

The run indexes were searched for arXiv:2508.08017, the source title, Alberti–Marchese, flat-chain conjecture, finite-dimensional Banach spaces, metric currents, and normal-current restriction language. Web searches on June 19, 2026 for the same finite-dimensional Banach transfer phrases found the Alberti–Marchese paper and adjacent metric-current literature, but no direct finite-dimensional Banach statement.

REFERENCES

- D. Bate, E. Caputo, J. Takac, P. Valentine, P. Wald, *Structure of Metric 1-currents: approximation by normal currents and representation results*, arXiv:2508.08017.
- G. Alberti and A. Marchese, *On the structure of flat chains with finite mass*, arXiv:2311.06099.